

Extension: Contextual Bandits

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Extends the MVP bandits (§5 of `mvp_specification.tex`) by conditioning action selection on activity level, and adds a decaying-epsilon variant.

1 Formulation

Context $c_t = s_t.\text{activity} \in \{\text{sedentary}, \text{active}\}$. Each algorithm maintains separate parameters per (c, a) pair:

- **Thompson Sampling:** $(\alpha_{c,a}, \beta_{c,a})$ posteriors. Sample $\theta_{c_t,a} \sim \text{Beta}(\alpha_{c_t,a}, \beta_{c_t,a})$, pick $a^* = \arg \max_a \theta_{c_t,a}$.
- **Epsilon-Greedy:** $Q_{c,a}$ values. With probability ϵ explore, otherwise $a^* = \arg \max_a Q_{c_t,a}$.
- **UCB:** $a^* = \arg \max_a [Q_{c_t,a} + c\sqrt{\ln(N_{c_t})/n_{c_t,a}}]$ where $N_{c_t} = \sum_a n_{c_t,a}$.
- **Decaying Epsilon-Greedy:** Linear decay: $\epsilon_t = \max(\epsilon_{\min}, \epsilon_{\text{start}}(1 - t/T_{\text{decay}}))$.

2 Optimal Policy

Nudge when sedentary ($0.3 > 0.1$), idle when active ($0.6 > 0.5$). Contextual bound: $E[r] = 3/7 \approx 0.429$ per step, 192.9 total (14.3% above non-contextual optimal).

3 Hyperparameter Optimisation

Grid search over key parameters (50 seeds, 450 timesteps, both standard and contextual variants):

EG: $\epsilon \in [0.01, 0.50]$, $\Delta 0.05$
DEC: $\epsilon_{\text{start}} \in [0.1, 1.0]$, $\Delta 0.1$; $T_{\text{decay}} \in [25, 700]$, $\Delta 25$
UCB: $c \in [0.1, 10.0]$, $\Delta 0.5$

$\epsilon = 0.05$ vs $\epsilon = 0.10$: identical means (158.1), but $\epsilon = 0.10$ has 11% lower variance.

4 Results

50 seeds, 450 timesteps. Optimised hyperparameters: EG $\epsilon = 0.05$, DEC $\epsilon_{\text{start}} = 0.2$, $T_{\text{decay}} = 200$, $\epsilon_{\min} = 0.01$, UCB $c = 0.5$.

Contextual TS and UCB reach the per-context optimal (0.429) by the last 50 steps. Standard UCB benefits most from hyperparameter search (-23.1% to -15.2%). DEC front-loads exploration,

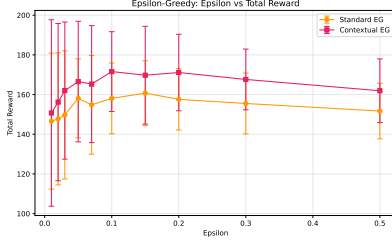


Figure 1: *
EG: ϵ vs reward

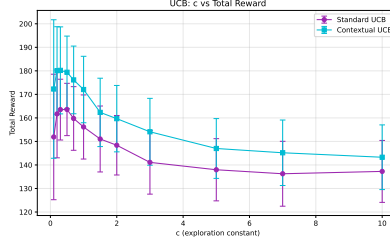


Figure 2: *
UCB: c vs reward

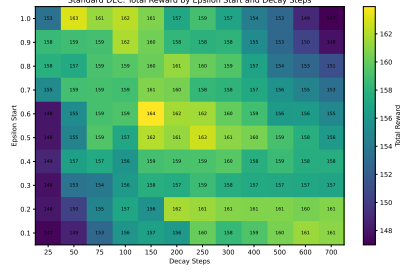


Figure 3: *
DEC: heatmap

Figure 4: Hyperparameter sensitivity (50 seeds, ± 1 SD). EG optimum $\epsilon = 0.15$, UCB optimum $c = 0.5$ (both interior). DEC optimum at $\epsilon_{\text{start}} = 1.0$ — the theoretical maximum, since $\Pr(\text{explore}) = \min(1, \epsilon)$.

Agent	Total Reward	Per Step	Last 50 Steps	vs Contextual Opt
Contextual TS	178.7 ± 15.8	0.397	0.429	0.0%
Contextual UCB	179.4 ± 15.4	0.399	0.430	0.0%
Contextual EG	166.6 ± 30.4	0.370	0.405	−13.6%
Contextual DEC	167.0 ± 35.2	0.371	0.396	−13.4%
Standard UCB	163.6 ± 11.1	0.364	0.382	−15.2%
Standard DEC	161.7 ± 18.4	0.359	0.375	−16.2%
Standard TS	159.6 ± 12.7	0.355	0.373	−17.2%
Standard EG	158.1 ± 19.9	0.351	0.375	−18.0%
Contextual optimal	192.9	0.429	0.429	—
Non-contextual optimal	168.8	0.375	0.375	—

achieving 161.7 vs EG’s 158.1 in the standard setting; the contextual advantage disappears as per-context Q-values already provide sufficient exploration.

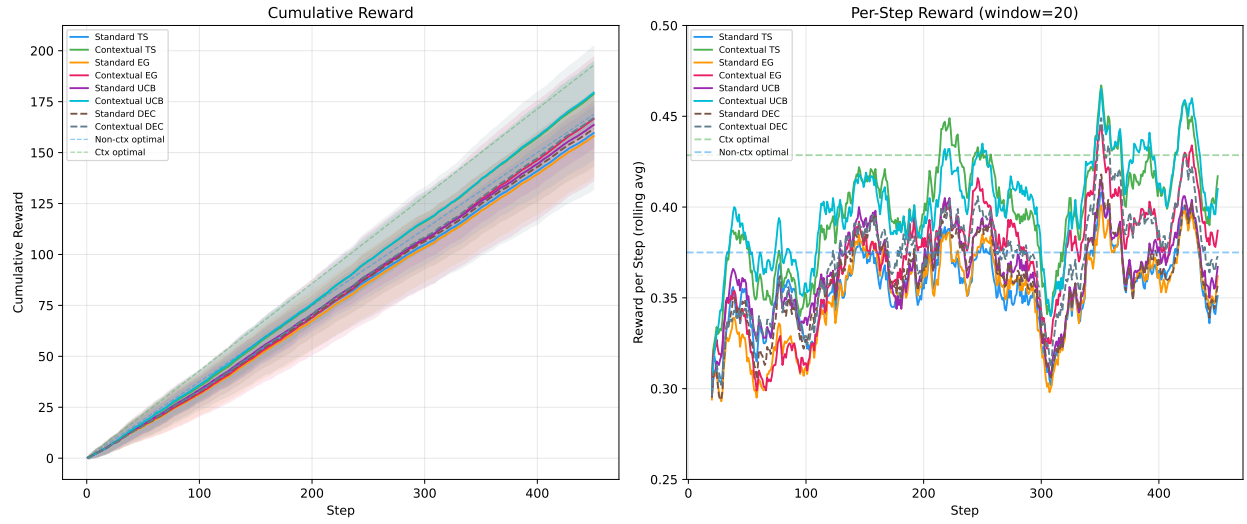


Figure 5: Cumulative reward and per-step reward (50 seeds, ± 1 SD). Contextual agents converge toward $3/7 \approx 0.429$; standard agents plateau lower.